

CHAPTER 9

Custom Exception

1. throw keyword

throw is used to explicitly throw an exception.

Syntax:

```
throw new ExceptionName();
```

Example:

```
public class ThrowExample {  
    public static void main(String[] args) {  
        int age = 15;  
  
        if (age < 18) {  
            throw new  
ArithmeticException("You are not eligible to  
vote");  
        }  
  
        System.out.println("Eligible to vote");  
    }  
}
```

}

2. throws keyword

throws is used in a method declaration to indicate that a method may produce an exception.

Syntax:

```
returnType      methodName()      throws  
ExceptionName {  
}
```

Example:

```
import java.io.FileNotFoundException;  
  
public class ThrowsExample {  
  
    static      void      checkFile()      throws  
FileNotFoundException {
```

```

        throw                                     new
FileNotFoundException("File not found");
    }

    public static void main(String[] args) {
        try {
            checkFile();
        } catch (FileNotFoundException e) {

            System.out.println(e.getMessage());
        }
    }
}

```

Difference between throw and throws
throw:

Used to explicitly throw an exception

Used to declare possible exceptions

Used inside a method

Used in the method declaration

throws:

Throws one exception at a time

Can declare multiple exceptions

Followed by an exception object

Followed by exception class names

```
void test() throws Exception { // declares  
    throw new Exception();      // actually  
throws  
}
```

Custom Exception for Age Validation

```
class InvalidAgeException extends Exception {  
  
    InvalidAgeException(String message) {  
        super(message);  
    }  
}
```

```
public class CustomExceptionDemo {  
  
    public static void main(String[] args) {  
  
        int age = 16;  
  
        try {  
            if (age < 18) {
```

```
        throw new
InvalidAgeException(
        "Age must be 18 or
above!"
    );
}

System.out.println("You are
eligible.");

    } catch (InvalidAgeException e) {
        System.out.println("Exception: " +
e.getMessage());
    }
}
}
```

Exception: Age must be 18 or above!

Custom Exception is a user-defined exception created by extending the Exception class. It is used to handle specific errors based on application requirements.